

Slides for the Stochastic Calculus Course

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Gaussian vectors

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Gaussian vectors

Let $(\Omega, \mathcal{A}, P)$ be a probability space.

Definition

[0.3.1] Let $X = (X_1, \dots, X_n)$ be a random vector of $\mathbb{R}^n$, X is said to be a Gaussian vector if for all $x = (x_1, \dots, x_n)$ in $\mathbb{R}^n$, the real random variable $\langle x, X \rangle$ is Gaussian.

Example

If $(X_1, \dots, X_n)$ are independent Gaussian random variables, then, $X = (X_1, \dots, X_n)$ is a Gaussian vector.

Gaussian vectors

Proposition

[0.3.1] If X is a Gaussian vector of $\mathbb{R}^n$ then, $\forall x \in \mathbb{R}^n$,

$$\Phi_X(x) = e^{i\langle x, m \rangle} e^{-\frac{x^t \Sigma x}{2}}$$

where

$$m = (E[X_1], \dots, E[X_n])$$

and

$$\Sigma = [\text{cov}(X_i, X_j)]_{1 \leq i, j \leq n}.$$

In this case, we say that X follows a $\mathcal{N}(m, \Sigma)$.

Corollary

[0.3.1] (cf. prop 0.2.1) If $X = (X_1, X_2)$ is a Gaussian vector of $\mathbb{R}^2$ then

$$X_1 \perp X_2 \Leftrightarrow \text{cov}(X_1, X_2) = 0.$$

Exercise

[0.3.1] Let $(Z, X_1, \dots, X_n)$ be a Gaussian vector such that $\forall 1 \leq i \leq n, Z \perp X_i$. Show that $Z \perp (X_1, \dots, X_n)$.

Exercise

[0.3.2] Let $G = (G_1, \dots, G_n)$ be a n -sample of a $\mathcal{N}(0, 1)$, $m \in \mathbb{R}^n$ and A be a $n \times n$ matrix. Show that $m + AG$ follows a $\mathcal{N}(m, AA^t)$. Propose an algorithm to simulate an arbitrary Gaussian vector (cf. exercise 0.2.1).

Proposition

[0.3.2] If X follows a $\mathcal{N}(m, \Sigma)$ and if Σ is nonsingular, then, X is absolutely continuous with respect to the Lebesgue measure on $\mathbb{R}^n$ with density

$$\frac{1}{(2\pi)^{\frac{n}{2}} (\det(\Sigma))^{\frac{1}{2}}} e^{-\frac{1}{2}(x-m)^t \Sigma^{-1}(x-m)}.$$

If Σ is singular, there exist $(c_1, \dots, c_n) \in \mathbb{R}^n - \{(0, \dots, 0)\}$ such that

$$c_1 X_1 + \dots + c_n X_n = \text{cste} \text{ with probability one.}$$

Conditional expectation

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Conditional expectation

Let $(\Omega, \mathcal{A}, P)$ be a probability space, X a random variable in $L^1(\Omega, \mathcal{A}, P)$ and $\mathcal{G}$ a sub σ -algebra of $\mathcal{A}$.

Definition

(Theorem) **[0.4.1]** *There exists a random variable $Z \in L^1$ such that*

i) Z is $\mathcal{G}$ -measurable

ii) $E[XU] = E[ZU] \forall U$ $\mathcal{G}$ -measurable and bounded.

Z is denoted by $E[X|\mathcal{G}]$ and is called the conditional expectation of X given $\mathcal{G}$. Moreover, Z is unique up to a.s equality.

Conditional expectation

Let Y be a random variable.

Definition

[0.4.2] When $X \in L^1$, $E[X|Y]$ is defined to be $E[X|\sigma(Y)]$.

Remarque

[0.4.3] The preceding definition may be rewritten in the following way

- i) $E[X|Y]$ is $\sigma(Y)$ -measurable
- ii) $E[Xg(Y)] = E[E[X|Y]g(Y)] \quad \forall g \text{ borelian and bounded.}$

Classical properties of an expectation

Proposition

[0.4.1] Let X and Y be two random variables in L^1 and $\mathcal{G}$ a sub σ -algebra of $\mathcal{A}$.

a) (positivity) If $X \geq 0$ P -a.s., $E[X|\mathcal{G}] \geq 0$ P -a.s.

b) (linearity) If $(\alpha, \beta) \in \mathbb{R}^2$, $E[\alpha X + \beta Y|\mathcal{G}] = \alpha E[X|\mathcal{G}] + \beta E[Y|\mathcal{G}]$ P -a.s.

c) (monotony) If $X \geq Y$ P -a.s., $E[X|\mathcal{G}] \geq E[Y|\mathcal{G}]$ P -a.s. (Consider $\{E[Y|\mathcal{G}] - E[X|\mathcal{G}] \geq \varepsilon\}$.)

d) (Beppo-lévy) If X_n is a sequence of non negative random variables in L^1 with $X_n \uparrow X$ P -a.s., then $E[X_n|\mathcal{G}] \uparrow E[X|\mathcal{G}]$ P -a.s.

e) (Fatou) If X_n is a sequence of non negative random variables in L^1 , then $E[\liminf_n X_n|\mathcal{G}] \leq \liminf_n E[X_n|\mathcal{G}]$ P -a.s.

f) (Dominated convergence) If X_n is a sequence of random variables in L^1 such that $X_n \xrightarrow[a.s.]{} X$ and $\forall n \in \mathbb{N}$, $X_n \leq X \in L^1$, then, $E[X_n|\mathcal{G}] \xrightarrow[a.s. \text{ and } L^1]{} E[X|\mathcal{G}]$.

g) (Jensen) If $\psi : \mathbb{R} \rightarrow \mathbb{R}$ is a convex function such that $\psi(X) \in L^1$, then, $\psi(E[X|\mathcal{G}]) \leq E[\psi(X)|\mathcal{G}]$ P -a.s.

Proposition

[0.4.2] Let X and Y be two random variables in L^1 and $\mathcal{G}$ a sub σ -algebra of $\mathcal{A}$.

a) $E[E[X|\mathcal{G}]] = E[X]$

b) If X is $\mathcal{G}$ -measurable, $E[X|\mathcal{G}] = X$ P -a.s.

c) (Taking out what is known) If Y is $\mathcal{G}$ -measurable and bounded, $E[XY|\mathcal{G}] = YE[X|\mathcal{G}]$ P -a.s.

d) (Role of independence) If $\sigma(X) \perp \mathcal{G}$, $E[X|\mathcal{G}] = E[X]$ P -a.s.

e) (Tower property) If $\mathcal{G}'$ is a sub σ -algebra of $\mathcal{A}$ such that $\mathcal{G}' \subset \mathcal{G}$, then, $E[E[X|\mathcal{G}]|\mathcal{G}'] = E[X|\mathcal{G}']$ P -a.s.

An important result

Proposition

[0.4.3] *If $\sigma(X) \perp \mathcal{G}$ and if Y is $\mathcal{G}$ -measurable, then, for all borelian function $\Phi : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that $E[|\Phi(X, Y)|] < \infty$, one has*

$$E[\Phi(X, Y)|\mathcal{G}] = \psi(Y) \text{ where } \psi(y) = E[\Phi(X, y)].$$

Proposition

[0.4.4] *Let $(Z, X_1, \dots, X_n)$ be a Gaussian vector. Then, there exist real numbers $(a, b_1, \dots, b_n)$ such that*

$$E[Z|X_1, \dots, X_n] = a + \sum_{i=1}^n X_i b_i.$$

Stochastic processes

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Definition

[0.5.1] A stochastic process on $(\Omega, \mathcal{A}, P)$, indexed by an arbitrary set $T \subset \mathbb{R}_+$, is a collection $(X_t)_{t \in T}$ of random variables on $(\Omega, \mathcal{A}, P)$ with values on a space E (for us, $E = \mathbb{R}$ and T is an interval).

Stochastic processes

Definition

[0.5.2] Two stochastic processes $(X_t)_{t \in T}$ and $(Y_t)_{t \in T}$ are equidistributed if, $\forall n \in \mathbb{N}^*, \forall (t_1, \dots, t_n) \in T^n, (X_{t_1}, \dots, X_{t_n}) \stackrel{\mathcal{D}}{=} (Y_{t_1}, \dots, Y_{t_n})$.

Definition

[0.5.3] A stochastic process $(X_t)_{t \in T}$ is a version of the process $(Y_t)_{t \in T}$ if, $\forall t \in T, X_t = Y_t$ P - a.s.

Definition

[0.5.4] Two stochastic processes $(X_t)_{t \in T}$ and $(Y_t)_{t \in T}$ are indistinguishable when $P(\{\omega | \forall t \in T, X_t(\omega) = Y_t(\omega)\}) = 1$

Rk: These definitions are more and more restrictive.

Stochastic processes

Definition

[0.5.5] We say that a stochastic process is continuous if, for P almost all ω , $t \in T \rightarrow X_t(\omega)$ is continuous.

Exercise

Let $(X_t)_{t \in T}$ and $(Y_t)_{t \in T}$ be stochastic processes.

1) Supposing that $(X_t)_{t \in T}$ and $(Y_t)_{t \in T}$ are continuous, show that if $(X_t)_{t \in T}$ is a version of $(Y_t)_{t \in T}$, then they are indistinguishable.

2) On the probability space $([0, 1], \mathcal{B}([0, 1]), dx)$, let $(X_t)_{t \in \mathbb{R}_+}$ be the process defined by $X_t(\omega) = \omega + at$ with $a > 0$ and $(Y_t)_{t \in \mathbb{R}_+}$ the process defined by $Y_t(\omega) = \omega + at$ if $t \neq \omega$ and $Y_t(\omega) = 0$ otherwise. Show that $(X_t)_{t \in \mathbb{R}_+}$ is a version of $(Y_t)_{t \in \mathbb{R}_+}$ but that they are not indistinguishable.

Definition

[0.5.6] A stochastic process $(X_t)_{t \in \mathbb{R}_+}$ is measurable if the function

$$(t, \omega) \in (\mathbb{R}_+, \mathcal{B}(\mathbb{R}_+)) \times (\Omega, \mathcal{A}) \rightarrow X_t(\omega) \in (\mathbb{R}, \mathcal{B}(\mathbb{R}))$$

is measurable i.e $\forall T \in \mathbb{R}_+, \forall E \in \mathcal{B}(\mathbb{R})$,

$$\{(t, \omega); 0 \leq t \leq T, X_t(\omega) \in E\} \in \mathcal{B}([0, T]) \otimes \mathcal{A}.$$

Rk: If $(X_t)_{t \in \mathbb{R}_+}$ is measurable and $f \in C_b(\mathbb{R}, \mathbb{R})$, $Y_t = \int_0^t f(X_s) ds$ a random variable.

Exercise

Show that a continuous process is measurable.

Definition

[0.5.7] Classically we denote

$$L^2(\Omega \times [0, T]) = \left\{ (X_t)_{t \in [0, T]} \text{ measurable; } E \left[\int_0^T X_s^2 ds \right] < \infty \right\}$$

the Hilbert space of squared integrable stochastic processes.

Stochastic processes

Let $(\Omega, \mathcal{A}, P)$ be a probability space.

Definition

[0.5.8] A non decreasing collection $(\mathcal{F}_t)_{t \in \mathbb{R}_+}$ of sub σ -algebras of $\mathcal{A}$ is called a filtration. This filtration is said to be complete if $\forall t \in \mathbb{R}_+, \mathcal{N} \subset \mathcal{F}_t$ where

$$\mathcal{N} = \{N \subset \Omega; \exists A \in \mathcal{A}, N \subset A, P(A) = 0\}.$$

Remarque

[0.5.3] We will always suppose that a filtration is complete changing $\mathcal{F}_t$ into $\sigma(\mathcal{N} \cup \mathcal{F}_t)$.

Definition

[0.5.9] A stochastic process $(X_t)_{t \in \mathbb{R}_+}$ is adapted to the filtration $(\mathcal{F}_t)_{t \in \mathbb{R}_+}$ if X_t is $\mathcal{F}_t$ measurable.

Remarque

[0.5.4] A stochastic process $(X_t)_{t \in \mathbb{R}_+}$ is always adapted to its natural filtration $\mathcal{F}_t^X = \sigma(X_s; 0 \leq s \leq t)$.

Remarque

[0.5.5] *There are several advantages to work with complete filtrations, in particular,*

- a) if $X_t \stackrel{\text{a.s.}}{=} Y_t$ and if X_t is $\mathcal{F}_t$ measurable $\Rightarrow Y_t$ is $\mathcal{F}_t$ measurable (thus, any version of an adapted process is an adapted process).*
- b) if $X_t^n \xrightarrow{\text{a.s.}} X_t$ and if $\forall n \geq 0, X_t^n$ is $\mathcal{F}_t$ measurable $\Rightarrow$ then X_t is $\mathcal{F}_t$ measurable.*

Definition

[0.5.10] A stochastic process $(X_t)_{t \in \mathbb{R}_+}$ is progressively measurable if, $\forall T > 0$ the function

$$(t, \omega) \in ([0, T], \mathcal{B}([0, T])) \times (\Omega, \mathcal{F}_T) \rightarrow X_t(\omega) \in (\mathbb{R}, \mathcal{B}(\mathbb{R}))$$

is measurable.

Remarque

[0.5.6] A progressively measurable process is measurable and adapted.

Stochastic processes

Exercise

For $0 \leq t_1 \leq \dots \leq t_n = T$, let F_{t_i} be a $\mathcal{F}_{t_i}$ -measurable random variable and

$$X_t = \sum_{i=1}^{n-1} F_{t_i} 1_{[t_i, t_{i+1}[}(t). \quad (1)$$

Show that $(X_t)_{t \in [0, T]}$ is progressively measurable. We will denote $\mathcal{E}([0, T] \times \Omega)$ for the elements of the form (2) with $F_{t_i} \in L^2(\mathcal{F}_{t_i})$.

Exercise

When $(X_t)_{t \in \mathbb{R}_+}$ is progressively measurable show that the process $(Y_t)_{t \in \mathbb{R}_+}$ where $Y_t = \int_0^t f(X_s) ds$ ($f \in C_b(\mathbb{R}, \mathbb{R})$) is adapted (Use Fubini's theorem).

Proposition

[0.5.1] A continuous process $(X_t)_{t \in \mathbb{R}_+}$ is progressively measurable when it is adapted.

Definition

[0.5.11] We denote

$$L^2_{prog}(\Omega \times [0, T]) = \left\{ (X_t)_{t \in [0, T]} \text{ prog meas; } E \left[\int_0^T X_s^2 ds \right] < \infty \right\}.$$

Theorem

[0.5.2] The space $L^2_{prog}(\Omega \times [0, T])$ equipped with its natural norm is complete. Moreover, $\mathcal{E}([0, T] \times \Omega)$ is dense in $L^2_{prog}(\Omega \times [0, T])$.

Properties of the BM

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Definition

Standard Brownian motion (B.M) is a stochastic process $(B_t)_{t \in [0, T]}$ fulfilling :

a) $B_0 = 0$ P-a.s.

b) B is continuous i.e $t \rightarrow B_t(w)$ is continuous for P almost all w .

c) B has independent increments: For Si $t > s$, $B_t - B_s$ is independent of $\mathcal{F}_s^B = \sigma(B_u, u \leq s)$.

d) the increments of B are stationary and gaussian: For $t > s$, $B_t - B_s$ follows a $\mathcal{N}(0, t - s)$.

MARTINGALE PROPERTIES

Proposition

[1.3.1] *The stochastic processes (B_t) , $((B_t)^2 - t)$ and $(e^{\theta B_t - \theta^2 \frac{t}{2}})$ ($\theta \in \mathbb{R}$) are $(\mathcal{F}_t^B)_{t \in \mathbb{R}_+}$ martingales.*

Proposition

[1.3.2] *Let $(X_t)_{t \geq 0}$ be a continuous martingale (with respect to the filtration $(\mathcal{F}_t^X)_{t \geq 0}$) starting from 0. Then X is a B.M if and only if the process $t \rightarrow e^{\theta X_t - \theta^2 \frac{t}{2}}$ is a martingale for all $\theta \in \mathbb{R}$.*

Proposition

[1.3.3] We put $B_t^{(s)} = B_{t+s} - B_s$, for fixed s , $Y_t = cB_{\frac{t}{c^2}}$, $c > 0$, $Z_t = tB_{\frac{1}{t}}$, $t > 0$, $Z_0 = 0$. Then, the stochastic processes $-B_t$, $B_t^{(s)}$, Y_t and Z_t are standard Brownian motions.

REGULARITY OF THE TRAJECTORIES

Proposition

[1.3.4] Let (B_t) be a Brownian motion. Then P -a.s,

$$a) \limsup_{t \rightarrow +\infty} \frac{B_t}{\sqrt{t}} = \limsup_{t \rightarrow 0+} \frac{B_t}{\sqrt{t}} = +\infty$$

$$b) \liminf_{t \rightarrow +\infty} \frac{B_t}{\sqrt{t}} = \liminf_{t \rightarrow 0+} \frac{B_t}{\sqrt{t}} = -\infty$$

Corollary

[1.3.1] P -a.s, the Brownian sample path (B_t) is nowhere differentiable from the right (resp. nowhere differentiable from the left).

Proposition

[1.3.5]

$$P(\{\omega; t \rightarrow B_t(\omega) \text{ is monotone in no interval}\}) = 1$$

VARIATIONS/QUADRATIC VARIATIONS

Proposition

[1.3.6] For $t > 0$, we set $\forall n \in \mathbb{N}, \forall j \in \{0, \dots, 2^n\}, t_j^n = \frac{tj}{2^n}$. Then,

$$Z_t^n = \sum_{j=1}^{2^n} |B_{t_j^n} - B_{t_{j-1}^n}|^2 \xrightarrow{\text{a.s. and } L^2} t.$$

Corollary

[1.3.2]

$$\sum_{j=1}^{2^n} |B_{t_j^n} - B_{t_{j-1}^n}| \xrightarrow{\text{a.s.}} +\infty.$$

The Brownian paths are a.s. of infinite variation on any interval.

Proposition

[1.3.7] For any $f : \mathbb{R} \rightarrow \mathbb{R}$ measurable and bounded, for $s < t$,

$$E[f(B_t)|\mathcal{F}_s^B] = E[f(B_t)|B_s] = \frac{1}{\sqrt{2\pi(t-s)}} \int_{\mathbb{R}} f(y) e^{-\frac{(y-B_s)^2}{2(t-s)}} dy.$$

GEOMETRIC BROWNIAN MOTION

Definition

[1.3.1] For $(b, \sigma) \in \mathbb{R}^2$, the stochastic process

$$X_t = X_0 e^{(b - \frac{1}{2}\sigma^2)t + \sigma B_t}$$

is called a *geometric Brownian motion*.

Exercise 1.3.1: a) Show that $X_t e^{-bt}$ is a martingale.

b) Show that $E[X_t|X_0] = X_0 e^{bt}$ and $\text{Var}(X_t|X_0) = X_0^2 e^{2bt} (e^{\sigma^2 t} - 1)$.

c) For $f \in C_b(\mathbb{R}, \mathbb{R})$ and $t > s$ show that

$$E[f(X_t)|\mathcal{F}_s^B] = \int_{-\infty}^{+\infty} f(X_s e^{(b - \frac{1}{2}\sigma^2)(t-s) + \sigma y \sqrt{t-s}}) \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy.$$

(the geometric Brownian motion is a Markov process)

Wiener Integral (see the construction of the Brownian Motion)

Let $(g_n)_{n \in \mathbb{N}}$ be i.i.d standard Gaussian random variables. We consider the Hilbert space $H = L^2([0, T], dx)$ and $(\chi_n)_{n \in \mathbb{N}}$ an associated orthonormal basis. We set, $\forall f \in H$,

$$I(f) = \sum_{n=0}^{\infty} \int_0^T f(s) \chi_n(s) ds g_n \quad (1)$$

and

$$I(f)_t = \sum_{n=0}^{\infty} \int_0^t f(s) \chi_n(s) ds g_n. \quad (2)$$

Wiener Integral (see the construction of the Brownian Motion)

Proposition

[1.3.9] Properties of the Wiener integral

- a) $f \in H \rightarrow \int_0^t f(s)dB_s$ is *linear*.
- b) $(\int_0^t f(s)dB_s)_{t \in [0, T]}$ is a *continuous* and *centered Gaussian* process with covariance function $\Gamma(s, t) = \int_0^{\inf(s, t)} f^2(u)du$.
- c) $(\int_0^t f(s)dB_s)_{t \in [0, T]}$ is *adapted* with respect to $(\mathcal{F}_t^B)_{t \in [0, T]}$ with *independent increments* (but *no stationarity*).
- d) When $(f, g) \in H^2$, $E \left[\int_0^t f(s)dB_s \int_0^u g(s)dB_s \right] = \int_0^{\inf(t, u)} f(s)g(s)ds$.
- e) $\left(\int_0^t f(s)dB_s \right)_{t \in [0, T]}$ and $\left(\left(\int_0^t f(s)dB_s \right)^2 - \int_0^t f^2(s)ds \right)_{t \in [0, T]}$ are $(\mathcal{F}_t^B)_{t \in [0, T]}$ *martingales*.
- f) $(\int_0^t f(s)dB_s)_{t \in [0, T]}$ fulfills the *Markov property*.

Wiener Integral (see the construction of the Brownian Motion)

Proposition

[1.3.10] When $f \in C^1([0, T], \mathbb{R})$, then, $\forall T \geq t \geq 0$,

$$\int_0^t f(s)dB_s = f(t)B_t - \int_0^t f'(s)B_s ds.$$

Integrands of the form $f(B_t)$

We want to give sense to

$$\int_0^t f(B_s) dB_s$$

Proposition

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable with bounded derivative. Then, the serie

$$Z_n = \sum_{i=0}^{n-1} f(B_{\frac{it}{n}})(B_{\frac{(i+1)t}{n}} - B_{\frac{it}{n}}) \quad (3)$$

converges in L^2 and we denote by $\int_0^t f(B_s) dB_s$ its limit (in general, this limit doesn't follow a normal distribution).

Example: Compute $\int_0^t B_s dB_s$.

Ito formula for the Brownian motion

Proposition

Consider $f \in C^2(\mathbb{R}, \mathbb{R})$ with **bounded second derivative**. Then, $\forall t \in [0, T]$,

$$f(B_t) = f(B_0) + \int_0^t f'(B_s) dB_s + \frac{1}{2} \int_0^t f''(B_s) ds \quad P - as.$$

Often, we will use the following differential notation

$$df(B_s) = f'(B_s) dB_s + \frac{1}{2} f''(B_s) ds.$$

Sketch of the proof: Discretization, increments, Taylor expansion, Riemann integral and quadratic variations!

Stochastic integral

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The stochastic integral on $\mathcal{E}([0, T] \times \Omega)$

Consider

$$X_t = \sum_{i=1}^{n-1} F_{t_i} \mathbf{1}_{[t_i, t_{i+1}[}(t) \quad (1)$$

in $\mathcal{E}([0, T] \times \Omega)$ (one has $0 = t_1 \leq \dots \leq t_n = T$ and $F_{t_i} \in L^2(\mathcal{F}_{t_i})$).

Definition

[2.1.1] In the case of an elementary integrand, the stochastic integral is given by

$$\int_0^T X_s dB_s = \sum_{i=1}^{n-1} F_{t_i} (B_{t_{i+1}} - B_{t_i}).$$

Remark: For $0 \leq t \leq T$ one defines naturally

$$\int_0^t X_s dB_s = \int_0^T X_s \mathbf{1}_{[0, t]}(s) dB_s = \sum_{i=1}^{n-1} F_{t_i} (B_{(t_{i+1} \wedge t)} - B_{(t_i \wedge t)}). \quad (2)$$

When X is a step function (F_{t_i} is constant) we recover the Wiener integral but, in general, $\int_0^t X_s dB_s$ is no more Gaussian.

Properties of the Ito integral on $\mathcal{E}([0, T] \times \Omega)$

Proposition

- [2.1.1] a) $X \mapsto \int_0^t X_s dB_s$ is *linear*.
b) The process $(\int_0^t X_s dB_s)_{t \in [0, T]}$ is *continuous*.
c) $(\int_0^t X_s dB_s)_{t \in [0, T]}$ is an *adapted* process with respect to $(\mathcal{F}_t^B)_{t \in [0, T]}$.
d) $E \left[\int_0^t X_s dB_s \right] = 0$ and $\text{Var} \left(\int_0^t X_s dB_s \right) = E \left[\int_0^t X_s^2 ds \right]$.
e) For $0 \leq s \leq t \leq T$,

$$E \left[\int_s^t X_u dB_u \middle| \mathcal{F}_s^B \right] = 0 \quad \text{and} \quad E \left[\left(\int_s^t X_u dB_u \right)^2 \middle| \mathcal{F}_s^B \right] = E \left[\int_s^t X_u^2 du \middle| \mathcal{F}_s^B \right]. \quad (3)$$

- f) $(\int_0^t X_s dB_s)_{t \in [0, T]}$ is a *continuous and square integrable martingale* with respect to $(\mathcal{F}_t^B)_{t \in [0, T]}$ moreover

$$E \left[\sup_{0 \leq t \leq T} \left| \int_0^t X_s dB_s \right|^2 \right] \leq 4E \left[\int_0^T X_u^2 du \right]. \quad (4)$$

Extension to $L^2_{prog}(\Omega \times [0, T])$

Proposition

[2.2.1] Let $X \in L^2_{prog}(\Omega \times [0, T])$. If Φ_n and Φ'_n are two sequences of elements of $\mathcal{E}([0, T] \times \Omega)$ that converge toward X in $L^2_{prog}(\Omega \times [0, T])$ then

$$\lim \int_0^\cdot \Phi_n(., s) dB_s \underset{M^2([0, T])}{=} \lim \int_0^\cdot \Phi'_n(., s) dB_s.$$

Definition

If $X \in L^2_{prog}(\Omega \times [0, T])$ and if Φ_n is a sequence in $\mathcal{E}([0, T] \times \Omega)$ that converges toward X in $L^2_{prog}(\Omega \times [0, T])$, we denote $(\int_0^\cdot X_s dB_s)$ the limit of $(\int_0^\cdot \Phi_n(., s) dB_s)$ in $M^2([0, T])$

Properties of the Ito integral on $L^2_{\text{prog}}(\Omega \times [0, T])$

Proposition

- [2.2.2]** a) $X \mapsto \int_0^t X_s dB_s$ is *linear*.
b) The process $(\int_0^t X_s dB_s)_{t \in [0, T]}$ is *continuous*.
c) $(\int_0^t X_s dB_s)_{t \in [0, T]}$ is an *adapted* process with respect to $(\mathcal{F}_t^B)_{t \in [0, T]}$.
d) $E \left[\int_0^t X_s dB_s \right] = 0$ and $\text{Var} \left(\int_0^t X_s dB_s \right) = E \left[\int_0^t X_s^2 ds \right]$.
e) For $0 \leq s \leq t \leq T$,

$$E \left[\int_s^t X_u dB_u \middle| \mathcal{F}_s^B \right] = 0 \quad \text{and} \quad E \left[\left(\int_s^t X_u dB_u \right)^2 \middle| \mathcal{F}_s^B \right] = E \left[\int_s^t X_u^2 du \middle| \mathcal{F}_s^B \right]. \quad (5)$$

- f) $(\int_0^t X_s dB_s)_{t \in [0, T]}$ is a *continuous and square integrable martingale* with respect to $(\mathcal{F}_t^B)_{t \in [0, T]}$ moreover

$$E \left[\sup_{0 \leq t \leq T} \left| \int_0^t X_s dB_s \right|^2 \right] \leq 4E \left[\int_0^T X_u^2 du \right]. \quad (6)$$

Definition

[2.3.1] An Ito process is a continuous and adapted process on $[0, T]$ of the form

$$X_t = X_0 + \int_0^t \psi_s ds + \int_0^t \phi_s dB_s \quad (7)$$

where ϕ and ψ belong to $L^2_{\text{prog}}(\Omega \times [0, T])$ and $X_0 \in L^2(\mathcal{F}_0)$. We will adopt the following differential notation

$$dX_s = \psi_s ds + \phi_s dB_s.$$

Proposition

[2.3.1] The decomposition (7) is **unique** up to stochastic equivalence.

Corollary

[2.3.1] According to proposition 0.5.5, an Ito process is a martingale if and only if the part in $un ds (\psi)$ is equal to zero .

Stochastic integral w.r.t Ito processes

Definition

[2.3.2] Naturally we can extend the notion of stochastic integral with respect to an Ito process. If X is of the form (7), then, for $\theta \in L^2_{prog}(\Omega \times [0, T])$ fulfilling $\theta\psi \in L^2_{prog}(\Omega \times [0, T])$ and $\theta\phi \in L^2_{prog}(\Omega \times [0, T])$ one defines $\int_0^t \theta_s dX_s$ by

$$\int_0^t \theta_s dX_s = \int_0^t \theta_s \psi_s ds + \int_0^t \theta_s \phi_s dB_s. \quad (8)$$

Proposition

[2.3.2] Let X be an Ito process of the form (7), if $f \in C^2(\mathbb{R}, \mathbb{R})$ with bounded derivatives, then

$$f(X_t) = f(X_0) + \int_0^t f'(X_s) dX_s + \frac{1}{2} \int_0^t f''(X_s) \phi_s^2 ds. \quad (9)$$

Proposition

[2.3.3] Let X be an Ito process of the form (7), if $f \in C^{1,2}([0, T] \times \mathbb{R}, \mathbb{R})$ with bounded derivatives, then

$$f(t, X_t) = f(0, X_0) + \int_0^t f'_x(s, X_s) dX_s + \frac{1}{2} \int_0^t f''_{xx}(s, X_s) \phi_s^2 ds + \int_0^t f'_t(s, X_s) ds. \quad (10)$$

Extended Ito Calculus

The assumption $X \in L^2_{prog}(\Omega \times [0, T])$ made in order to construct stochastic integral is sometime **too restrictive**.

Define

$$\mathcal{H}^2_{loc}(\Omega \times [0, T]) = \left\{ (X_t)_{t \in [0, T]} \text{ prog meas; } \int_0^T X_s^2 ds < \infty P - a.s \right\}$$

and

$$\mathcal{H}^1_{loc}(\Omega \times [0, T]) = \left\{ (X_t)_{t \in [0, T]} \text{ prog meas; } \int_0^T |X_s| ds < \infty P - p.s \right\}.$$

Obviously we have

$$L^2_{prog}(\Omega \times [0, T]) \subset \mathcal{H}^2_{loc}(\Omega \times [0, T]) \subset \mathcal{H}^1_{loc}(\Omega \times [0, T]).$$

Extended Ito Calculus

Proposition

[2.4.1] *We can extend the stochastic integral to integrands in $\mathcal{H}_{loc}^2(\Omega \times [0, T])$. But, in this case, the stochastic process $\int_0^t X_s dB_s$ **is not in general a martingale** (e.g $E[\int_0^t X_s dB_s]$ may be different from zero). Nevertheless, the properties a), b) et c) on proposition 2.2.2 remain valid.*

Definition

[2.4.1] *A generalized Ito process is an adapted and continuous process $[0, T]$ of the form*

$$X_t = X_0 + \int_0^t \psi_s ds + \int_0^t \phi_s dB_s \quad (11)$$

where $\phi \in \mathcal{H}_{loc}^2(\Omega \times [0, T])$, $\psi \in \mathcal{H}_{loc}^1(\Omega \times [0, T])$ and X_0 is $\mathcal{F}_0^B$ measurable. We will use the shorthand differential notation

$$dX_s = \psi_s ds + \phi_s dB_s.$$

Proposition

[2.4.2] *The decomposition (11) is unique:*

$$X_t = X_0 + \int_0^t \psi_s ds + \int_0^t \phi_s dB_s = X'_0 + \int_0^t \psi'_s ds + \int_0^t \phi'_s dB_s \quad (12)$$

implies

$$X_0 = X'_0 \text{ } P - \text{a.s.}, \psi = \psi' \text{ } dx \otimes P - \text{a.s.}, \phi = \phi' \text{ } dx \otimes P - \text{a.s.}$$

Proposition

[2.4.3] *If (X_t) is a martingale of the form (11) then $\psi = 0 \text{ } dx \otimes P - \text{a.s.}$*

Extended Ito Formula

Proposition

[2.4.4] Let X be an Ito process of the form (11). If $f \in C^2(\mathbb{R}, \mathbb{R})$, then,

$$f(X_t) = f(X_0) + \int_0^t f'(X_s) dX_s + \frac{1}{2} \int_0^t f''(X_s) \phi_s^2 ds. \quad (13)$$

where

$$\int_0^t f'(X_s) dX_s = \int_0^t f'(X_s) \psi_s ds + \int_0^t f'(X_s) \phi_s dB_s.$$

Proposition

[2.4.5] Let X be an Ito process of the form (11). If $f \in C^{(1,2)}([0, T] \times \mathbb{R}, \mathbb{R})$, then,

$$f(t, X_t) = f(0, X_0) + \int_0^t f'_x(s, X_s) dX_s + \frac{1}{2} \int_0^t f''_{xx}(s, X_s) \phi_s^2 ds + \int_0^t f'_t(s, X_s) ds. \quad (14)$$

Extended Ito Formula

Proposition

[2.4.6] Let X be an Ito process of the form (11) such that $\forall t \in [0, T]$, $X_t \in \Theta$ P -a.s. If $f \in C^2(\Theta, \mathbb{R})$, then,

$$f(X_t) = f(X_0) + \int_0^t f'(X_s) dX_s + \frac{1}{2} \int_0^t f''(X_s) \phi_s^2 ds. \quad (15)$$

where

$$\int_0^t f'(X_s) dX_s = \int_0^t f'(X_s) \psi_s ds + \int_0^t f'(X_s) \phi_s dB_s.$$

Exercise: (Integration by parts formula)

Let X and Y be two Ito processes $[0, T]$ of the form

$$X_t = X_0 + \int_0^t \psi_s ds + \int_0^t \phi_s dB_s$$

and

$$Y_t = Y_0 + \int_0^t \psi'_s ds + \int_0^t \phi'_s dB_s.$$

Integration by parts

Exercise: (Integration by parts formula)

Let X and Y be two Ito processes $[0, T]$ of the form

$$X_t = X_0 + \int_0^t \psi_s ds + \int_0^t \phi_s dB_s$$

and

$$Y_t = Y_0 + \int_0^t \psi'_s ds + \int_0^t \phi'_s dB_s.$$

Show that

$$X_t Y_t = X_0 Y_0 + \int_0^t Y_s dX_s + \int_0^t X_s dY_s + \int_0^t \phi_s \phi'_s ds. \quad (17)$$

Proposition

[2.5.1] For $(b, \sigma) \in \mathbb{R}^2$, there exists a unique (in the sense of definition 0.5.4) Ito process (S_t) such that

$$dS_t = bS_t dt + \sigma S_t dB_t$$

(with $S_0 = x_0$). This process is given by

$$S_t = x_0 e^{(b - \frac{1}{2}\sigma^2)t + \sigma B_t}. \quad (18)$$

SDE: Introduction to the general case

A stochastic differential equation (SDE) is an equation of the form

$$(E) \begin{cases} X_0 = x_0 \in \mathbb{R} \\ dX_t = b(t, X_t)dt + \sigma(t, X_t)dB_t \end{cases}$$

where

$$b : \mathbb{R}_+ \times \mathbb{R} \rightarrow \mathbb{R}$$

$$\sigma : \mathbb{R}_+ \times \mathbb{R} \rightarrow \mathbb{R}.$$

Definition

[2.5.1] A solution to (E) is a continuous stochastic process $(X_t)_{t \in [0, T]}$ adapted to the Brownian filtration such that

- $\int_0^T |b(s, X_s)| + |\sigma(s, X_s)|^2 ds < \infty$ a.s
- $\forall t \in [0, T],$

$$X_t = x_0 + \int_0^t b(s, X_s) ds + \int_0^t \sigma(s, X_s) dB_s \text{ a.s.}$$

Theorem

[2.5.1] Under the hypothesis

$$\mathcal{H}_1 \left\{ \begin{array}{l} \exists K > 0, \forall t \in [0, T], \forall (x, y) \in (\mathbb{R}^n)^2 \\ |b(t, x) - b(t, y)| + |\sigma(t, x) - \sigma(t, y)| \leq K|x - y| \\ |b(t, x)| + |\sigma(t, x)| \leq K(1 + |x|) \end{array} \right.$$

equation (E) has a unique solution $(X_t)_{t \in [0, T]}$ fulfilling

$$E \left[\sup_{s \in [0, T]} X_s^2 \right] < \infty.$$

$\mathcal{H}_1$ is sufficient but not necessary cf

$$(\text{CIR}): \quad dr_t = a(b - r_t)dt + \sigma\sqrt{r_t}dB_t$$

avec $(a, b, \sigma, r_0) \in \mathbb{R}_+$.

Euler Scheme for ODE

We consider the following ordinary differential equation (EDO):

$$y'(t) = f(t, y(t)); y(0) = x_0.$$

When an explicit solution does not exist we can approximate on $[0, T]$:

- We consider a subdivision **subdivision**, here, $\{t_0 = 0, t_1 = \frac{T}{N}, \dots, t_N = T\}$.
- We approximate the solution at the points of our subdivision

$$\hat{y}(0) = x_0$$

$$\hat{y}(t_k) = \hat{y}(t_{k-1}) + \frac{T}{N} f(t_{k-1}, \hat{y}(t_{k-1})).$$

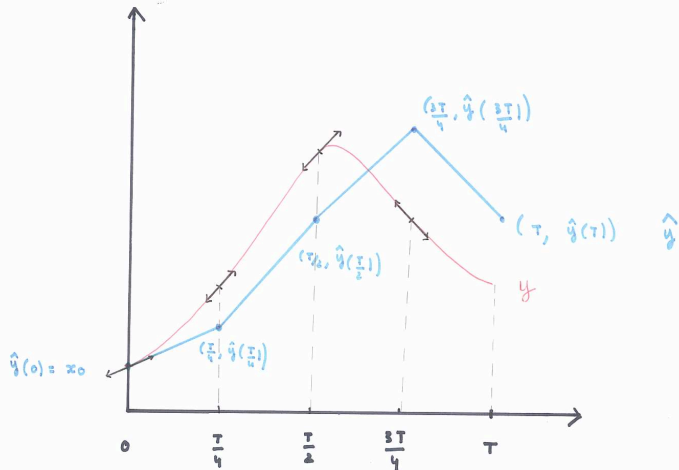
- We approximate the solution on $[0, T]$ by the piecewise linear interpolation of $(t_k, \hat{y}(t_k))_{0 \leq k \leq N}$.

Under weak hypotheses on f

$$\|y - \hat{y}\|_{\infty} \xrightarrow[N \rightarrow \infty]{} 0$$

Euler Scheme for ODE

Schema d'Euler Pour que $m = 4$



Euler Scheme for SDE

In finance we often have to compute quantities of the form

$$E[f(X_T)]$$

where $(X_t)_{t \in \mathbb{R}_+}$ is solution of (E) .

Using Monte Carlo methods

$$E[f(X_T)] \approx \frac{1}{M} \sum_{i=1}^M f(X_T^i).$$

Problem:

- We don't know in general the **distribution** of (X_T)
- We can't simulate in **continuous time** $\Rightarrow$ **Discretization**

Definition

For $N \in \mathbb{N}^*$,

- We consider a *subdivision*, here, $\{t_0 = 0, t_1 = \frac{T}{N}, \dots, t_N = T\}$.
- We approximate the solution at the points of our subdivision using

$$\hat{X}^N(0) = x_0$$

$$\hat{X}^N(t_k) = \hat{X}^N(t_{k-1}) + \frac{T}{N} b(t_{k-1}, \hat{X}^N(t_{k-1})) + \sigma(t_{k-1}, \hat{X}^N(t_{k-1}))(B_{t_k} - B_{t_{k-1}}).$$

- We approximate the solution on $[0, T]$ by the piecewise linear interpolation of $(t_k, \hat{X}^N(t_k))_{0 \leq k \leq N}$:

$$\forall t \in [t_k, t_{k+1}],$$

$$\hat{X}^N(t) = \hat{X}^N(t_k) + (t - t_k)b(t_k, \hat{X}^N(t_k)) + \sigma(t_k, \hat{X}^N(t_k))(B_t - B_{t_k}).$$

Two technical results

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The Girsanov theorem

In this chapter $(B_t)_{t \in [0, T]}$ is a standard B.M defined on a probability space $(\Omega, \mathcal{A}, P)$. Moreover, this probability space is equipped with the natural brownian filtration $(\mathcal{F}_t^B)_{t \in [0, T]}$. We denote by E the expectation under P .

Reminder: We have already proved (exercise 0.6.1) the following result: If X is a random variable that follows, under a probability P , a $\mathcal{N}(m, \sigma^2)$ then, under the probability Q (equivalent to P) having the density

$$L = e^{-\frac{mX}{\sigma^2}} e^{+\frac{m^2}{2\sigma^2}}, \quad (1)$$

with respect to P , X follows a $\mathcal{N}(0, \sigma^2)$.

Here the aim is to extend this result in a dynamical way

Girsanov theorem

If $m \in \mathbb{R}$, we consider the stochastic process $(\tilde{B}_t)_{t \in [0, T]}$ defined by

$$\tilde{B}_t = B_t + mt. \quad (2)$$

$(\tilde{B}_t)_{t \in [0, T]}$ Brownian motion under the probability $P \Leftrightarrow m = 0$.

Now we want to find a probability Q which makes $(\tilde{B}_t)_{t \in [0, T]}$ a standard Brownian motion

Girsanov theorem

Since $\tilde{B}_t$ is a $\mathcal{N}(mt, t)$, according to (1) we define

$$L_t = e^{-m\tilde{B}_t} e^{\frac{m^2 t}{2}} = e^{-mB_t} e^{-\frac{m^2 t}{2}}. \quad (3)$$

Then, one has

- a)** for fixed t , $\tilde{B}_t$ is, under the probability Q_t having the density L_t with respect to P , a $\mathcal{N}(0, t)$,
- b)** the stochastic process $(L_t)_{t \in [0, T]}$ is a martingale under P (**exponential martingale**).

Lemma

[3.1.1] A stochastic process $(M_t)_{t \in [0, T]}$ is martingale under Q if and only if the process $(L_t M_t)_{t \in [0, T]}$ is a martingale under P .

Proposition

[3.1.1] If $m \in \mathbb{R}$. Under the probability Q defined by

$$\frac{dQ}{dP} = L_T,$$

$(\tilde{B}_t)_{t \in [0, T]}$ is a *standard Brownian motion*.

More generally, if $(\theta_t)_{t \in [0, T]} \in \mathcal{H}_{loc}^2(\Omega \times [0, T])$, one defines

$$L_t = e^{-\int_0^t \theta_s dB_s - \frac{1}{2} \int_0^t \theta_s^2 ds} \quad (4)$$

and using the Ito formula $dL_t = -L_t \theta_t dB_t$.

Theorem

(Girsanov) Let $(\theta_t)_{t \in [0, T]} \in \mathcal{H}_{loc}^2(\Omega \times [0, T])$ be such that the stochastic process $(L_t)_{t \in [0, T]}$ defined by

$$L_t = e^{-\int_0^t \theta_s dB_s - \frac{1}{2} \int_0^t \theta_s^2 ds} \quad (5)$$

is a martingale under P . Then, under the probability Q with density

$$\frac{dQ}{dP} = e^{-\int_0^T \theta_s dB_s - \frac{1}{2} \int_0^T \theta_s^2 ds},$$

$(\tilde{B}_t)_{t \in [0, T]}$ where $\tilde{B}_t = B_t + \int_0^t \theta_s ds$ is a standard Brownian motion.

Proposition

(Novikov) The process $(L_t)_{t \in [0, T]}$ is a martingale if

$$E[e^{\frac{1}{2} \int_0^T \theta_s^2 ds}] < \infty.$$

Martingale representation theorem

Theorem

(*Ito representation theorem*) let $F \in L^2(\Omega, \mathcal{F}_T^B, P)$, then there exists a unique $(\theta_t)_{t \in [0, T]} \in L^2_{prog}(\Omega \times [0, T])$ so that

$$F = E[F] + \int_0^T \theta_s dB_s. \quad (6)$$

Theorem

(*Martingale representation theorem*) Let $(M_t)_{t \in [0, T]}$ be a continuous and square integrable martingale (with respect to the brownian filtration), there exists a unique stochastic process $(\theta_t)_{t \in [0, T]} \in L^2_{prog}(\Omega \times [0, T])$ such that

$$M_t = E[M_0] + \int_0^t \theta_s dB_s. \quad (7)$$

Calcul sto IRFA: Introduction to Finance in continuous time

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Where the stochastic integral is coming from???

Let us consider a **time interval** $[0, T]$ and an associated **subdivision** $0 = t_1 < \dots t_n = T$.

We suppose that (**Bachelier**) the value of a financial product is given by B_t at time t .

We consider **a trader that adopts the following strategy**: for $f \in L^2([0, T])$, $\forall i \in \{1, \dots, n-1\}$

- a) he **buys** $f(t_i)$ assets at time t_i (the unit price is B_{t_i})
- b) he **sells** them at time t_{i+1} (the unit price is $B_{t_{i+1}}$)
- c) during the period $[t_i, t_{i+1}]$ the **profit** is $f(t_i)(B_{t_{i+1}} - B_{t_i})$.

On $[0, T]$ the **total profit** is given by

$$\sum_{i=1}^{n-1} f(t_i)(B_{t_{i+1}} - B_{t_i}).$$

Where the stochastic integral is coming from???

If the trading is done in **continuous time**, this profit becomes

$$\int_0^T f(s)dB_s.$$

Furthermore, let us consider a financial product whose price at time T (denoted by $P(T)$) is of the form

$$P(T) = k + \int_0^T f(t)dB_t. \quad (1)$$

If the financial market fulfills a natural hypothesis (**No Arbitrage opportunity**), it can be shown that:

- 1) the asset price at $t = 0$ is none other than k . (**PRICING**)
- 2) If the seller of such a product obtains k at time $t = 0$ he is able to **deliver** the asset at time T **independently of the variations of the market**. (**HEDGING**)

Modelling financial markets in continuous time

- Time scale $[0, T]$ (T **maturity date**)
- A set Ω (**macroeconomic configurations**) during this period.
- A σ -algebra $\mathcal{A}$ on Ω (**financial information**)
- A probability P called the **historical probability**.

In practice the information needed for the perfect knowledge of $\mathcal{A}$ and P **is not available**.

We are going to consider two types of financial assets.

The non risky asset

We suppose that during the period $[0, T]$ it is **possible to borrow and lend cash at a constant rate $r > 0$** .

If S_t^0 denotes the value at time t of one euro lent at time $t = 0$, one has,
 $\forall t \in [0, T]$,

$$S_t^0 = e^{rt}.$$

This asset is called the non-risky asset because its value is **independent from randomness**.

Actualization: $\tilde{X}_t = \frac{X_t}{e^{rt}}$

The risky asset

Risky: The value of this asset depends on **randomness** (unpredictable macro-economic changes).

Let us denote by $\mathbf{S}_t$ the (random) value of this asset at time t .

$(S_t)_{t \in [0, T]}$ is adapted to its natural filtration $(\mathcal{F}_t)_{t \in [0, T]}$ ($\mathcal{F}_t = \sigma(\mathbf{S}_u; u \leq t)$) and **supposed to be continuous**.

The σ -algebra $\mathcal{F}_t$ represents the **information** available at time t .

HYP: We suppose that the dynamic of the risky asset is given by an **Ito process** whose value at 0 is a constant!

No frictions

HYP: no market frictions

- i) There are no transaction costs or taxes.**
- ii) It is possible to short sell the risky asset and to borrow and lend cash.**
- iii) All assets are perfectly divisible.**
- iv) Trading is done in continuous time.**
- v) The risky asset does not pay a dividend.**

Definition (4.2.1)

A financial strategy is a **progressively measurable** process $(H_t = (\theta_t^0, \theta_t)_{t \in [0, T]})$ with values in $\mathbb{R}^2$ such that the integrals

$$\int_0^T \theta_t^0 dS_t^0 \quad \text{and} \quad \int_0^T \theta_t dS_t$$

are well defined.

We associate to any **financial strategy** a **financial portfolio** containing at time $t \in [0, T]$ θ_t^0 unities of non risky asset and θ_t unities of risky asset.

Thus, the value at time $t \in [0, T]$ of such a portfolio is given by

$$V_t^H = \theta_t^0 S_t^0 + \theta_t S_t \tag{2}$$

Self financed strategies

Definition (4.2.2)

A financial strategy $(H_t)_{t \in [0, T]}$ is said to be self-financed if the value of the associated portfolio fulfills the following S.D.E:

$$dV_t^H = \theta_t^0 dS_t^0 + \theta_t dS_t. \quad (3)$$

Lemma (4.2.1)

Let $(H_t)_{t \in [0, T]}$ and $(H'_t)_{t \in [0, T]}$ be two self-financed financial strategies such that $\forall t \in [0, T], \mathbf{V}_t^H = \mathbf{V}_t^{H'}$ P.a.s then H and H' are **equal up to indistinguishability**.

Proposition (4.2.1)

A strategy is **self-financed** if and only if

$$d\tilde{V}_t^H = \theta_t d\tilde{S}_t$$

Arbitrage opportunities (A.O)

Definition (4.2.3)

A self financed strategy $(H_t)_{t \in [0, T]}$ is said to be an *arbitrage opportunity* (A.O) if the following conditions are fulfilled

$$V_0^H = 0, V_T^H \geq 0 \text{ P - a.s and } P(V_T^H > 0) > 0. \quad (4)$$

Rk: Definition **independent of the choice of an equivalent probability.**

Equivalent martingale measure (EMM)

Definition (4.2.4)

An **E.M.M** is a probability measure P^* equivalent to P under which the actualized price $(\tilde{S}_t)_{t \in [0, T]}$ of the risky asset is a **martingale**.

Definition (4.2.5)

A financial strategy $(H_t)_{t \in [0, T]}$ is said to be P^* **admissible** if it is **self-financed** and if the actualized value of the associated portfolio (V_t^H) is a **martingale** under P^* .

Proposition (4.2.2)

There are **no arbitrage opportunities** (N.A.O) among P^* admissible financial strategies.

Definition (4.2.6)

***Contingent claims** are assets whose prices depend on the values of other assets (for us the risky one). In our framework, it will be $\mathcal{F}_T$ measurable random variables (fulfilling some technical integrability conditions).*

Examples: **European** type options.

Step 1: **Model the dynamic of the asset.**

We have to deal with two **contradictory constraints** :

- 1) The model has to be sufficiently fine to represent the reality.
- 2) The model has to be sufficiently simple to be operational (cf infra).

Study plan for CH5

Step 2: **Study the properties of the model.**

Notably, is the **N.A.O** condition fulfilled?

Step 3: **Propose a price for a large family of contingent claims (PRICING).**

Moreover, we have to be able to effectively perform this price:

- a) Obtaining **closed form formula** with known (at least from a statistical point of view) parameters.
- b) Obtaining theoretical formula that can be approximate by efficient **numerical methods** (discretization, Monte Carlo).

Step 4: Propose financial strategies to reduce all risks due to random fluctuations of the underlying (HEDGING).

When the price of a contingent claim is fixed, how to use the premium to ensure its delivery independently of randomness?

Study plan for CH5

Step 5: (Finally....) **Confront the model with reality.**

1) **Reality of financial markets**

Several contingent claims are quoted on derivatives markets. It is especially the case for call and put options. Thus we have to

- a) Compare our prices with the ones proposed by the market (calibrating the parameters to fit the data)
- b) Propose prices for claims traded on over the counter markets.

2) **Reality of other models**

We have to compare our model with the existing ones. Some criteria may be:

- a) Precision
- b) Complexity (computing time...)
- c) Field of applications (products that may be priced and hedged....)

Calcul sto IRFA: Black and Scholes model

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The model

In the Black-Scholes model, **the dynamic of the risky asset** is given by the following SDE:

$$\underline{dS_t = bS_t dt + \sigma S_t dB_t} \quad (1)$$

with initial condition $S_0 = x_0 > 0$.

We have (prop 2.5.1) that this SDE has a unique solution given by the **geometric Brownian motion**

$$S_t = x_0 e^{(b - \frac{1}{2}\sigma^2)t + \sigma B_t}. \quad (2)$$

Existence of an EMM

Proposition

[5.1.1] *In the Black and Scholes model **there exists** an EMM (def 4.2.4).*

Proposition

[5.3.1] *In the Black-Scholes there exists a **unique** E.M.M (i.e P^* given by **Girsanov Theorem**).*

Definition (4.2.1)

A financial strategy is a **progressively measurable** process $(H_t = (\theta_t^0, \theta_t)_{t \in [0, T]})$ with values in $\mathbb{R}^2$ such that the integrals

$$\int_0^T \theta_t^0 dS_t^0 \quad \text{and} \quad \int_0^T \theta_t dS_t$$

are **well defined**.

We associate to any **financial strategy** a **financial portfolio** containing at time $t \in [0, T]$ θ_t^0 unities of non risky asset and θ_t unities of risky asset.

Thus, the value at time $t \in [0, T]$ of such a portfolio is given by

$$V_t^H = \theta_t^0 S_t^0 + \theta_t S_t \tag{3}$$

Definition (4.2.1)

A financial strategy is a **progressively measurable** process $(H_t = (\theta_t^0, \theta_t))_{t \in [0, T]}$ with values in $\mathbb{R}^2$ such that the integrals

$$\int_0^T \theta_t^0 dS_t^0 \quad \text{and} \quad \int_0^T \theta_t dS_t$$

are **well defined**.

In **Black and Scholes model**,

$$\theta_0 \in \mathcal{H}_{loc}^1(\Omega \times [0, T]) \quad (4)$$

and

$$\theta \in \mathcal{H}_{loc}^2(\Omega \times [0, T]), \quad (5)$$

is sufficient to make the two integrals **well-defined**.

P^* admissible financial strategies

Definition

[5.2.1] In the Black-Scholes model, a financial strategy H is said to be P^* **admissible** if it is **self-financed** and if the actualized value of the associated portfolio is a squared integrable and nonnegative **martingale under P^*** .

The following result is none other than proposition 4.2.2.

Proposition

There is **N.A.O** among P^* admissible strategies.

P^* -completeness of the market

Definition

[5.2.2] *In this part, a contingent claim is a nonnegative random variable h in $L^2(\Omega, \mathcal{F}_T^B, P^*)$.*

Definition

[5.2.3] *A contingent claim is said to be P^* attainable if it is equal to the final value of a portfolio associated to a strategy P^* admissible.*

Definition

[5.2.4] *The financial market is P^* complete if all contingent claims are P^* attainable.*

Theorem

[5.2.1] *The Black and Scholes model is P^* complete. Moreover, the value at time t of **any** hedging portfolio for the contingent claim h is given by*

$$V_t = E^*[he^{-r(T-t)}|\mathcal{F}_t^B]. \quad (6)$$

The **brilliant idea** of Black and Scholes was to propose the following definition for the price of a contingent claim.

Definition

[5.4.2] The **price**, at time $t \in [0, T]$, of a contingent claim h is the value at time t of any hedging portfolio associated to an admissible strategy (such a portfolio always exists according to Theorem 5.2.1). The price process will be denoted by $(P_t^h)_{t \in [0, T]}$, it is a martingale under P^* .

$$P_t^h = E^*[he^{-r(T-t)} | \mathcal{F}_t^B].$$

Pricing and Hedging when $h = f(S_T)$

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a measurable function such that $E^*[f^2(S_T)] < +\infty$.
The value at time t of the contingent claim $h = f(S_T)$ is

$$P_t^h = E^*[e^{-r(T-t)} f(S_T) | \mathcal{F}_t^B]$$

thus

$$P_t^h = e^{-r(T-t)} \int_{-\infty}^{+\infty} f(S_t e^{(r-\frac{1}{2}\sigma^2)(T-t)+\sigma y\sqrt{T-t}}) \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy. \quad (7)$$

Thus $P_t^h = F(t, S_t)$ where

$$F(t, x) = e^{-r(T-t)} \int_{-\infty}^{+\infty} f(x e^{(r-\frac{1}{2}\sigma^2)(T-t)+\sigma y\sqrt{T-t}}) \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy. \quad (8)$$

Pricing and Hedging when $h = f(S_T)$

Lemma

[5.5.1] Under weak hypotheses (for example when h is the payoff of a Call or a Put), the function F belongs to $C^{1,2}([0, T] \times \mathbb{R}, \mathbb{R})$.

Proposition

[5.5.1] When $h = f(S_T)$, the quantity of risky asset to put in the hedging portfolio at time t is given by

$$\theta_t = \frac{\partial F}{\partial x}(t, S_t). \quad (9)$$

Moreover, the function F satisfies the following **PDE**:

$$\frac{\partial F}{\partial t}(t, x) + rx \frac{\partial F}{\partial x}(t, x) + \frac{\sigma^2 x^2}{2} \frac{\partial^2 F}{\partial x^2}(t, x) = rF(t, x) \quad (10)$$

with final condition $F(T, x) = h(x)$.

Black and Scholes formula

Proposition

[5.5.2] *The price at time t of an **European call** (with strike K and maturity T) is given by*

$$C_t = S_t N(d_1(t, S_t)) - Ke^{-r(T-t)} N(d_2(t, S_t)) \quad (11)$$

where

$$d_1(t, x) = \frac{\log(\frac{x}{K}) + (r + \frac{\sigma^2}{2})(T-t)}{\sigma\sqrt{T-t}} \text{ et } d_2(t, x) = \frac{\log(\frac{x}{K}) + (r - \frac{\sigma^2}{2})(T-t)}{\sigma\sqrt{T-t}} \quad (12)$$

and where N is the distribution function of a $\mathcal{N}(0, 1)$. In this case, the composition of the hedging portfolio is given by

$$\theta_t = N(d_1(t, S_t)) > 0 \text{ and } \theta_t^0 = -Ke^{-rT} N(d_2(t, S_t)) < 0. \quad (13)$$

Definition

We call “greeks” the following quantities:

- Δ measures the sensitivity of the price to the underlying

$$\Delta_t(S_t) = \frac{\partial F}{\partial x}(t, S_t) \quad (14)$$

- Γ measures the sensitivity of the delta to the underlying

$$\Gamma_t(S_t) = \frac{\partial^2 F}{\partial x^2}(t, S_t) \quad (15)$$

- Θ measures the sensitivity of the price to the time

$$\Theta_t(S_t) = \frac{\partial F}{\partial t}(t, S_t) \quad (16)$$

- ρ measures the sensitivity of the price to the interest rate

$$\rho_t(S_t) = \frac{\partial F}{\partial r}(t, S_t) \quad (17)$$

- vega (that is not a greek letter!!!) measures the sensitivity of the price to the volatility

$$\text{vega}_t(S_t) = \frac{\partial F}{\partial \sigma}(t, S_t). \quad (18)$$

The Greeks

	Call	Put
Δ	$N(d_1) > 0$	$-N(-d_1) < 0$
Γ	$\frac{1}{x\sigma\sqrt{T}}N'(d_1) > 0$	$\frac{1}{x\sigma\sqrt{T}}N'(d_1) > 0$
Θ	$-\frac{x\sigma}{2\sqrt{T}}N'(d_1) - Kre^{-rT}N(d_2) < 0$	$\frac{x\sigma}{2\sqrt{T}}N'(d_1) + Kre^{-rT}(N(d_2) - 1) ??$
ρ	$TKe^{-rT}N(d_2) > 0$	$TKe^{-rT}(N(d_2) - 1) < 0$
<i>vega</i>	$x\sqrt{T}N'(d_1) > 0$	$x\sqrt{T}N'(d_1) > 0$